Hyperbolas

Section 1: Standard Equations of Hyperbolas

1 Find the vertices and orientation of $\frac{x^2}{16} - \frac{y^2}{9} = 1$.

a. Horizontal, vertices at $(\pm 4, 0)$

b. Vertical, vertices at $(0, \pm 3)$

c. Horizontal, vertices at $(\pm 5, 0)$

d. Vertical, vertices at $(0, \pm 4)$

2 Find the vertices and orientation of $\frac{y^2}{25} - \frac{x^2}{144} = 1$.

a. Vertical, vertices at $(0, \pm 5)$

b. Horizontal, vertices at $(\pm 12, 0)$

c. Vertical, vertices at $(0, \pm 13)$

d. Horizontal, vertices at $(\pm 5, 0)$

3 Find the length of the transverse axis and conjugate axis for $\frac{(x-1)^2}{36} - \frac{(y+2)^2}{64} = 1$.

a. Transverse $= 12$, Conjugate $= 16$

b. Transverse $= 6$, Conjugate $= 8$

c. Transverse $= 16$, Conjugate $= 12$

d. Transverse $= 72$, Conjugate $= 128$

Section 2: Asymptotes of Hyperbolas

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4 State the equations of the asymptotes for the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

a. $y = \pm\frac{b}{a}x$

b. $y = \pm\frac{a}{b}x$

c. $y = \pm\frac{b^2}{a^2}x$

d. $y = \pm x$

5 Find the asymptotes of $\frac{x^2}{9} - \frac{y^2}{16} = 1$.

a. $y = \pm\frac{4}{3}x$

b. $y = \pm\frac{3}{4}x$

c. $y = \pm\frac{16}{9}x$

d. $y = \pm\frac{5}{3}x$

6 Find the asymptotes of the vertical hyperbola $\frac{y^2}{4} - \frac{x^2}{9} = 1$.

a. $y = \pm\frac{2}{3}x$

b. $y = \pm\frac{3}{2}x$

c. $y = \pm\frac{4}{9}x$

d. $y = \pm 2x$

Section 3: Foci and Eccentricity of Hyperbolas

7 What is the relationship between a, b, and focal distance c for a hyperbola?

a. $c^2 = a^2 + b^2$

b. $c^2 = a^2 - b^2$

c. $a^2 = b^2 + c^2$

d. $c = a + b$

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8 Find the foci of $\frac{x^2}{16} - \frac{y^2}{9} = 1$.

a. $(\pm 5, 0)$

b. $(0, \pm 5)$

c. $(\pm\sqrt{7}, 0)$

d. $(\pm 25, 0)$

9 Find the eccentricity $e = c/a$ of $\frac{x^2}{16} - \frac{y^2}{9} = 1$.

a. $5/4 = 1.25$

b. $4/5 = 0.8$

c. $3/4 = 0.75$

d. $\sqrt{7}/4$

Section 4: Identifying Conics from General Second-Degree Equations

10 For $Ax^2 + Cy^2 + Dx + Ey + F = 0$ ($A, C \neq 0$), which condition identifies a hyperbola?

a. A and C have opposite signs ($AC < 0$)

b. A and C have the same sign and $A = C$

c. A and C have the same sign and $A \neq C$

d. $A = 0$ or $C = 0$

11 Identify the conic: $9x^2 - 4y^2 - 36x + 8y - 4 = 0$.

a. Hyperbola

b. Ellipse

c. Parabola

d. Circle

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12 Convert $9x^2 - 4y^2 - 36x + 8y - 4 = 0$ to standard form.

a. $\frac{(x-2)^2}{4} - \frac{(y-1)^2}{9} = 1$

b. $\frac{(x-2)^2}{9} - \frac{(y-1)^2}{4} = 1$

c. $\frac{(y-1)^2}{9} - \frac{(x-2)^2}{4} = 1$

d. $(x-2)^2 - (y-1)^2 = 36$

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